

**ENHANCING ASTRONOMY LEARNING FOR GRADE 8
STUDENTS IN SRI LANKA THROUGH A STEM-
BASED AUGMENTED REALITY MOBILE
APPLICATION**

Project ID: 25-26J-418

Project Proposal Report

Shenali L.A.I – IT22896636

BSc (Hons) Degree in Information Technology Specializing in
Interactive Media

Department of Information Technology

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due acknowledgement is made in the text.

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Abstract

Teaching astronomy to middle school students often relies on static diagrams and verbal explanations that fail to convey the dynamic, spatial nature of celestial phenomena. This project proposes a low-cost, offline Augmented Reality (AR)–driven mobile learning application for Grade 8 students in Sri Lanka, integrating interactive 3D simulations, gamified quizzes, and teacher monitoring features. The system allows students to visualize and manipulate complex astronomical processes, including planetary motion, lunar phases, eclipses, and the Sun–Earth–Moon system, in real time. Developed using Unity, Vuforia, and ARCore, the application supports budget Android devices and offline functionality. Evaluation will include pre-/post-tests, achievement and motivation surveys, and teacher interviews to measure learning gains and usability. Expected outcomes include enhanced conceptual understanding, increased student engagement, and positive teacher feedback, providing a scalable and practical approach to integrating AR into STEM education.

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1. INTRODUCTION

Astronomy is a fundamental part of the middle school STEM curriculum, providing students with insights into the structure, dynamics, and evolution of the universe. However, teaching astronomy at the Grade 8 level often relies on static textbook diagrams and verbal explanations, which fail to adequately convey the dynamic and three-dimensional nature of celestial phenomena. Students frequently struggle to visualize concepts such as planetary motion, lunar phases, eclipses, and the Sun–Earth–Moon system, resulting in limited conceptual understanding and engagement [1], [2]. Traditional pedagogical approaches, although aligned with the curriculum, often lack interactivity and fail to address diverse learning styles, which can hinder the development of problem-solving skills and critical thinking in STEM education [2], [4].

Augmented Reality (AR) has emerged as a promising educational technology to bridge these gaps. By overlaying interactive 3D content onto real-world environments, AR enables learners to explore abstract scientific concepts in a more tangible and immersive manner [3], [4]. Several studies have demonstrated the positive impact of AR on learning outcomes in astronomy and science education. For example, Ferrari et al. [1] reported significant improvements in conceptual understanding among secondary school students using AR-based simulations. Similarly, Kasinathan et al. [5] developed the “First Discovery” AR application, which improved engagement and short-term knowledge retention among primary students through marker-based interactions and interactive 3D visualizations. Research conducted in Indonesia and Vietnam [2], [3] also indicates that AR can enhance students’ motivation, engagement, and problem-solving abilities when integrated with science curricula.

Despite the demonstrated benefits, most existing AR applications face several limitations. Many global AR solutions are not curriculum-aligned, lack offline functionality, or require high-end devices, limiting their accessibility in low-resource classrooms [2], [4]. Furthermore, few studies incorporate teacher perspectives, which are critical for

classroom integration and sustainable adoption of AR technologies. These gaps highlight the need for context-specific AR solutions that are low-cost, offline-compatible, curriculum-aligned, and designed with both students and teachers in mind.

To address these challenges, this project proposes the development of a marker-based AR mobile application tailored to the Sri Lankan Grade 8 astronomy syllabus. The system will enable students to scan textbook diagrams and interact with immersive 3D models, supplemented by gamified quizzes and a teacher dashboard for monitoring and guidance. Offline functionality and compatibility with budget Android devices ensure accessibility and scalability. The research methodology will combine requirement analysis, AR content design, system development, and iterative classroom testing. Effectiveness will be measured using student pre-/post-tests, achievement and motivation surveys, and teacher interviews to evaluate learning gains, engagement, and usability.

By integrating STEM education principles with immersive AR technology, this project aims to enhance conceptual understanding, promote active learning, and provide a practical, scalable solution for astronomy education in Sri Lankan classrooms. The proposed research addresses both technological and pedagogical gaps, offering an innovative approach to teaching complex scientific concepts while supporting teachers in effectively implementing AR-based learning.

1.1 Background and Literature Survey

Background

Astronomy is a core component of the middle school STEM curriculum, providing students with foundational knowledge about the universe, celestial bodies, and cosmic phenomena. Traditional teaching methods often rely on static textbook diagrams, verbal explanations, and teacher-led lectures. While these methods ensure curriculum coverage, they frequently fail to convey the dynamic, spatial, and three-dimensional aspects of astronomical concepts, such as planetary motion, lunar phases, eclipses, and the Sun–Earth–Moon system [1], [2]. Students often struggle to develop a deep conceptual understanding or retain complex information, leading to reduced engagement, motivation, and problem-solving skills [2].

Advances in technology, particularly Augmented Reality (AR), offer new opportunities to overcome these limitations. AR allows students to interact with virtual 3D models overlaid on real-world environments, providing immersive experiences that can enhance conceptual understanding, motivation, and engagement [3]. AR applications in education have shown promise in making abstract scientific concepts more tangible and interactive, bridging the gap between theoretical knowledge and visual-spatial comprehension.

Literature Survey

Several recent studies highlight the potential of AR in improving astronomy education:

Unlocking the Cosmos [1] investigated the efficacy of AR in secondary school astronomy instruction, reporting significant improvements in students' conceptual understanding and engagement compared to traditional methods. The study used pre-/post-tests with a large sample (~130 students), providing robust empirical evidence of AR's effectiveness.

Exploring Augmented Reality on Astronomy Education [2] focused on conceptual knowledge, motivation, and learning attitudes among Indonesian students. The findings

indicate that AR can improve both learning outcomes and students' interest in astronomy, although the study emphasized the importance of content alignment with the curriculum.

Enhancing Astronomy Skills in Vietnamese Middle Schools [3] demonstrated that mobile AR applications positively influence students' problem-solving skills, spatial reasoning, and active participation in classroom activities. The study also highlighted the benefits of marker-based AR for visualizing complex astronomical phenomena.

Development of Solar System Teaching Aids with AR [4] examined the integration of interactive 3D models and gamified learning in Indonesia. The study confirmed that AR-based tools enhance student engagement, though offline accessibility and low-end device compatibility were not extensively addressed.

First Discovery: Augmented Reality for Learning Solar Systems [5] presented a marker-based AR application for primary students, which increased engagement and short-term knowledge retention. While the target audience was younger, the study provides evidence that interactive AR can supplement traditional textbooks effectively.

1.2 Research gap

Over the past decade, Augmented Reality (AR) has been increasingly explored in educational contexts, particularly in STEM subjects. Prior studies demonstrate AR's potential to enhance visualization, motivation, and engagement in science learning [1], [2], [3], [5]. In astronomy education, interactive AR models allow students to explore dynamic phenomena such as planetary motion, lunar phases, and eclipses, improving conceptual understanding compared to traditional methods [1], [3], [5]. However, several limitations remain that restrict the widespread effectiveness and classroom integration of AR solutions.

1) Lack of Integrated, Curriculum-Aligned Systems

Many existing AR applications are limited in scope, focusing on either 3D visualization, simple quizzes, or engagement tracking, but rarely combining these features into a cohesive, curriculum-aligned system [2], [4]. Global AR tools like Star Walk 2 or Solar System AR are primarily designed for general audiences and are not adapted to specific national curricula, reducing their applicability in formal classroom settings [2], [4]. Furthermore, most solutions require high-end devices or continuous internet access, limiting accessibility in resource-constrained schools [4].

2) Limited Quantitative Evaluation of Learning Gains

Much of the current research emphasizes qualitative outcomes, such as student perceptions, motivation, and engagement, while neglecting rigorous quantitative validation of learning improvement [2], [5]. Although improvements in engagement are reported, few studies employ statistical methods to measure actual learning gains or compare AR interventions to traditional teaching approaches [1], [3]. Without robust experimental evidence, the true educational impact of AR in astronomy remains uncertain.

3) Insufficient Teacher Involvement

Teachers play a critical role in classroom adoption and sustained integration of AR technologies. However, most studies do not include teacher perspectives during design or evaluation [2], [5]. Lack of teacher involvement can result in AR tools that are difficult to integrate into lesson plans, reducing their effectiveness and long-term use in schools.

4) Absence of Offline, Low-Cost Solutions

Few AR applications are designed to work offline or on low-end devices, which is essential for deployment in developing countries like Sri Lanka [2], [4]. This technological constraint limits scalability and equitable access for students in schools with budget or connectivity limitations.

Research Contribution

This research aims to address these gaps by developing a **curriculum-aligned, offline AR mobile application** for Grade 8 astronomy that integrates interactive 3D simulations, gamified quizzes, and teacher dashboards. A robust experimental design will be employed, including pre-/post-tests and paired/independent t-tests, to provide statistically validated evidence of learning gains, engagement, and usability [1], [3], [5]. Teacher input will be incorporated throughout the design and evaluation process to ensure classroom feasibility and curriculum alignment.

Table 1 - system gap

Feature / System	Unlocking the Cosmos [1]	Exploring AR on Astronomy Education [2]	Enhancing Astronomy Skills (Vietnam) [3]	Solar System Teaching Aids (Indonesia) [4]	First Discovery [5]	Proposed System
Target Grade / Audience	Secondary (~Grade 8)	Middle school (Grades 7–8)	Middle school	Middle school	Primary (~ages 9–10)	Grade 8 (Sri Lanka)
Platform / Tools	Unity, marker-based AR	Unity / ARCore, marker-based AR	Unity, marker-based AR	Unity, AR simulations	Mobile marker-based AR	Unity + Vuforia + AR Foundation, marker-based AR
3D Interactive Simulations	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Gamified Quizzes	Limited / basic	Limited	Limited	✓ Yes	✓ Yes	✓ Yes, instant feedback with explanations
Curriculum Alignment	No	Partial	No	Partial	No	✓ Fully aligned with Sri Lankan Grade 8 syllabus
Offline Functionality	✗	✗	✗	✗	✗	✓ Works offline for low-resource schools
Teacher Dashboard / Support	✗	✗	✗	✗	✗	✓ Teacher-friendly dashboard, control features
Device Requirement	Medium–high	Medium	Medium	Medium	Low–medium	✓ Optimized for budget Android (≥ 2 GB RAM)
Evaluation / Validation	Pre-/post-tests	Surveys & perception studies	Pre-/post-tests	Limited evaluation	Engagement & usability	Pre-/post-tests, paired/independent t-tests, teacher interviews
Focus	Learning gains & engagement	Conceptual knowledge & motivation	Problem-solving skills	Visualization & quizzes	Engagement & short-term knowledge	Integrated, holistic AR learning for astronomy with assessment & teacher support

1.3 Research Problem

Astronomy education in Sri Lanka continues to rely predominantly on textbooks, diagrams, and lectures, which are insufficient for conveying the dynamic, three-dimensional nature of celestial phenomena. As a result, students often develop misconceptions and struggle with conceptual understanding, engagement, and problem-solving in astronomy. Although Augmented Reality (AR) has demonstrated global potential to enhance learning in STEM subjects, existing AR applications are primarily designed for general audiences, are not aligned with the national curriculum, overlook teacher involvement, and typically require high-end devices or continuous internet connectivity. These limitations restrict their adoption and effectiveness in Sri Lankan classrooms.

The core research problem is therefore: **How can a low-cost, offline, curriculum-aligned AR mobile application be designed and developed for Grade 8 astronomy to improve students' conceptual understanding, engagement, and problem-solving skills while supporting teachers in classroom integration?**

2. OBJECTIVES

2.1 Main Objective

The primary goal of this study is to design, develop, and evaluate a STEM-based Augmented Reality (AR) mobile application that enhances astronomy learning for Grade 8 students in Sri Lanka. The application targets abstract concepts such as planetary motion, lunar phases, and eclipses, providing an interactive and immersive experience to improve comprehension, engagement, and knowledge retention. Additionally, the study aims to assess the practical feasibility of classroom integration from the teachers' perspective.

2.2 Sub Objectives

- Identify teacher and student perceptions of current astronomy teaching methods

Gather insights on challenges, misconceptions, and attitudes toward digital learning to inform app design.
- Design an AR application aligned with the Grade 8 curriculum

Develop interactive 3D models, simulations, and activities that directly support national learning outcomes and classroom instruction.
- Evaluate application effectiveness using pre-test/post-test and paired t-tests

Measure student learning gains and provide quantitative evidence of the AR app's educational impact.
- Analyze teacher feedback on usability and classroom integration

Collect surveys and interviews to assess the app's practicality, user-friendliness, and alignment with lesson plans.
- Compare AR-based learning with traditional textbook instruction

Examine differences in engagement, understanding, and knowledge retention to support recommendations for technology-enhanced STEM learning.

3. METHODOLOGY

3.1 System Overview

The proposed system is a **comprehensive Augmented Reality (AR) learning platform** designed for Grade 8 astronomy students in Sri Lanka, integrating interactive 3D content, performance tracking, and teacher support. The system aims to bridge the gap between traditional textbook-based instruction and immersive, technology-enhanced learning.

The platform consists of **three interconnected components**:

1. Mobile Application (Front-End Interface):

- Provides students with access to interactive 3D simulations of celestial phenomena, including planetary motion, lunar phases, eclipses, and the Sun–Earth–Moon system.
- Offers gamified quizzes with instant feedback to reinforce learning and track student progress.
- Ensures a child-friendly, intuitive interface optimized for low-end Android devices.

2. AR Content Engine:

- Built using **Unity 3D**, **Vuforia**, and **AR Foundation** to deliver marker-based AR experiences.
- Handles real-time rendering of 3D models, AR interactions, and simulation logic.
- Allows students to manipulate virtual celestial objects to explore concepts dynamically and immersively

3. Database and Content Storage:

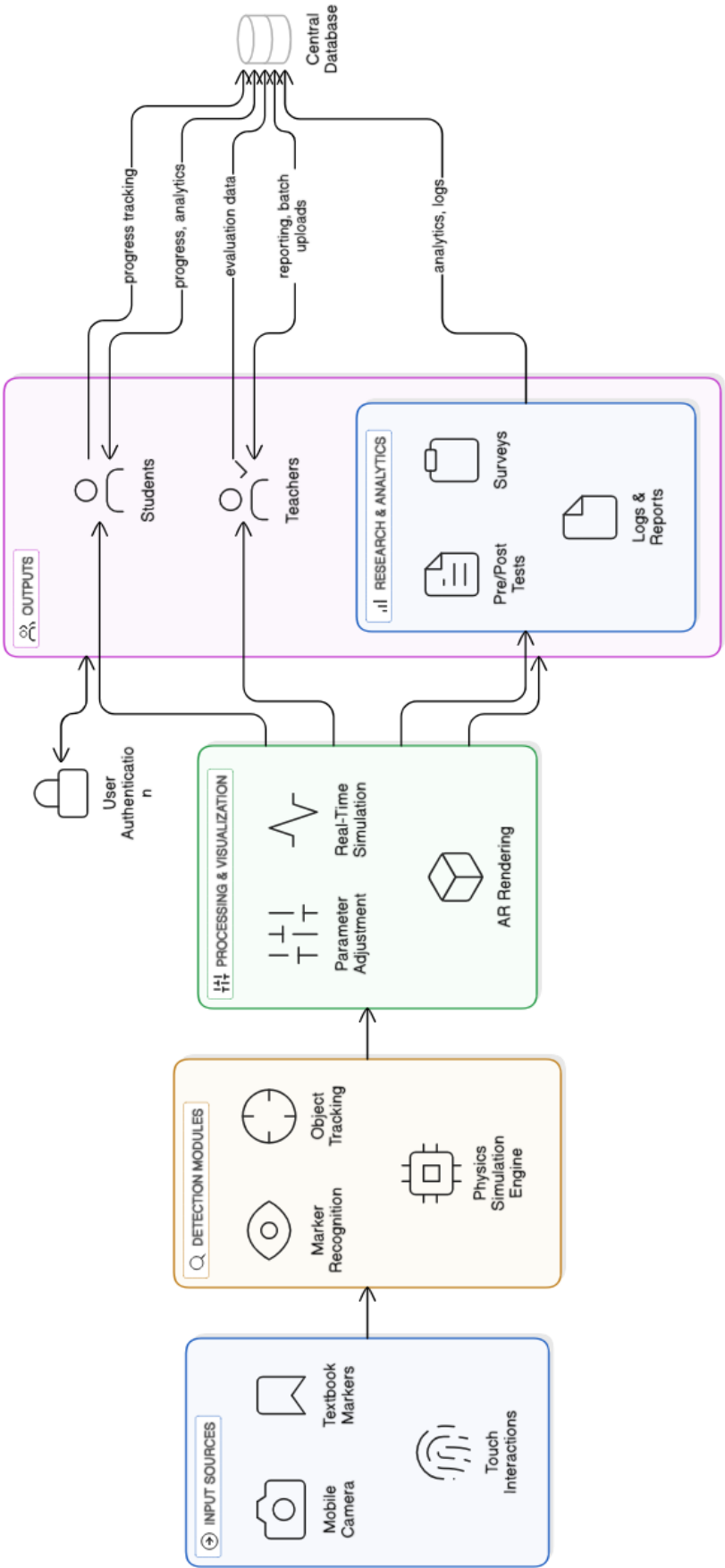
- Stores quiz results, student progress, teacher feedback, and AR content assets.
- Supports offline functionality, ensuring usability in classrooms with limited internet connectivity.
- Enables teacher monitoring and reporting via a dedicated dashboard, allowing instructional adjustments and personalized support.

3.2 System Workflow:

- Students scan textbook diagrams or AR markers using the mobile app, triggering the AR engine to display interactive 3D models.
- Learners interact with simulations and complete embedded quizzes, with performance data stored locally or synced to the database.
- Teachers access the dashboard to monitor class progress, control simulation visibility, and provide feedback, ensuring curriculum alignment and pedagogical relevance.

This integrated architecture ensures a **holistic, engaging, and scalable AR learning experience**, addressing the limitations of traditional methods while supporting teachers in the classroom.

Figure 1- system diagram



AR Astronomy Learning App - High Level Architecture

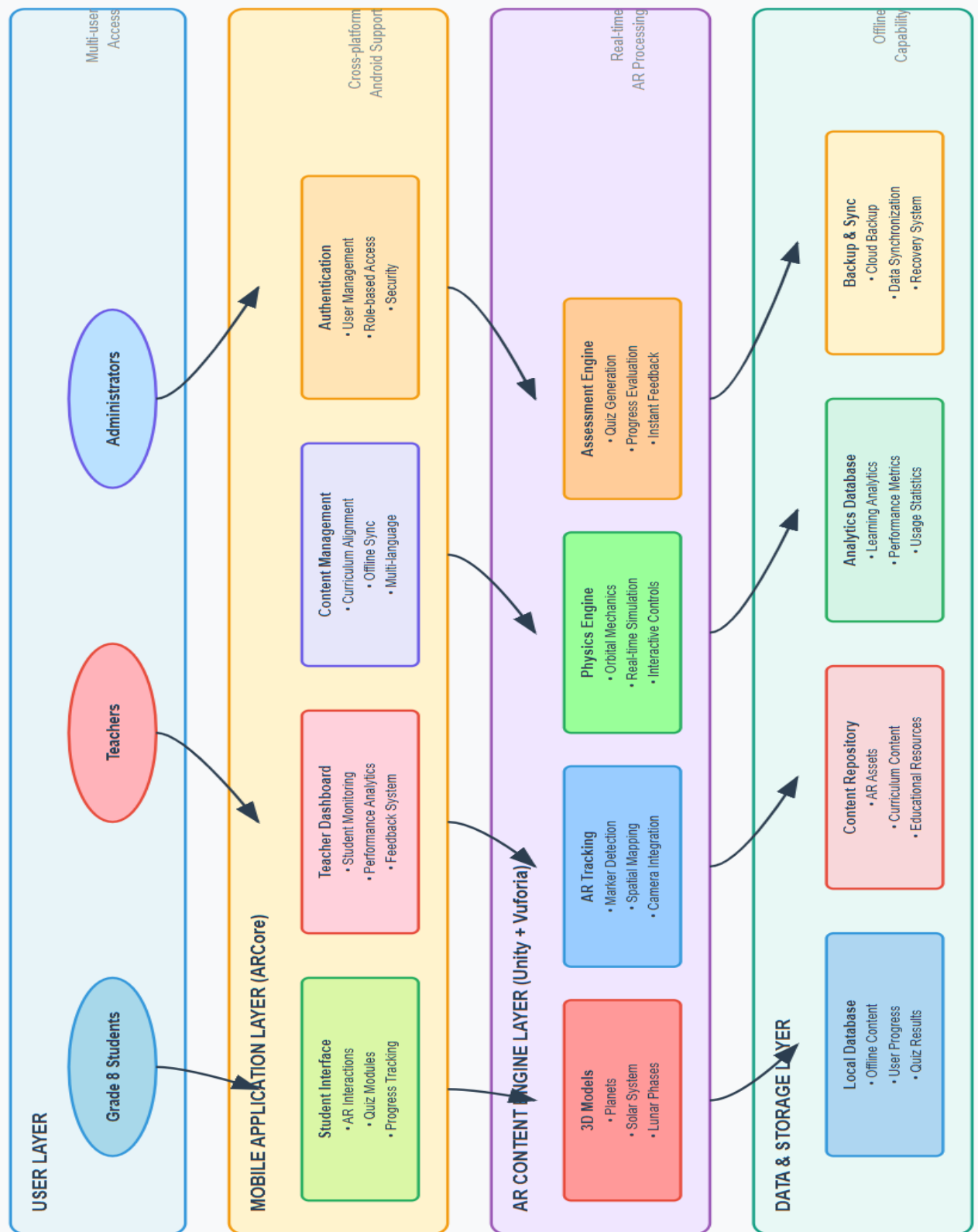


Figure 2- high lecture architcg design

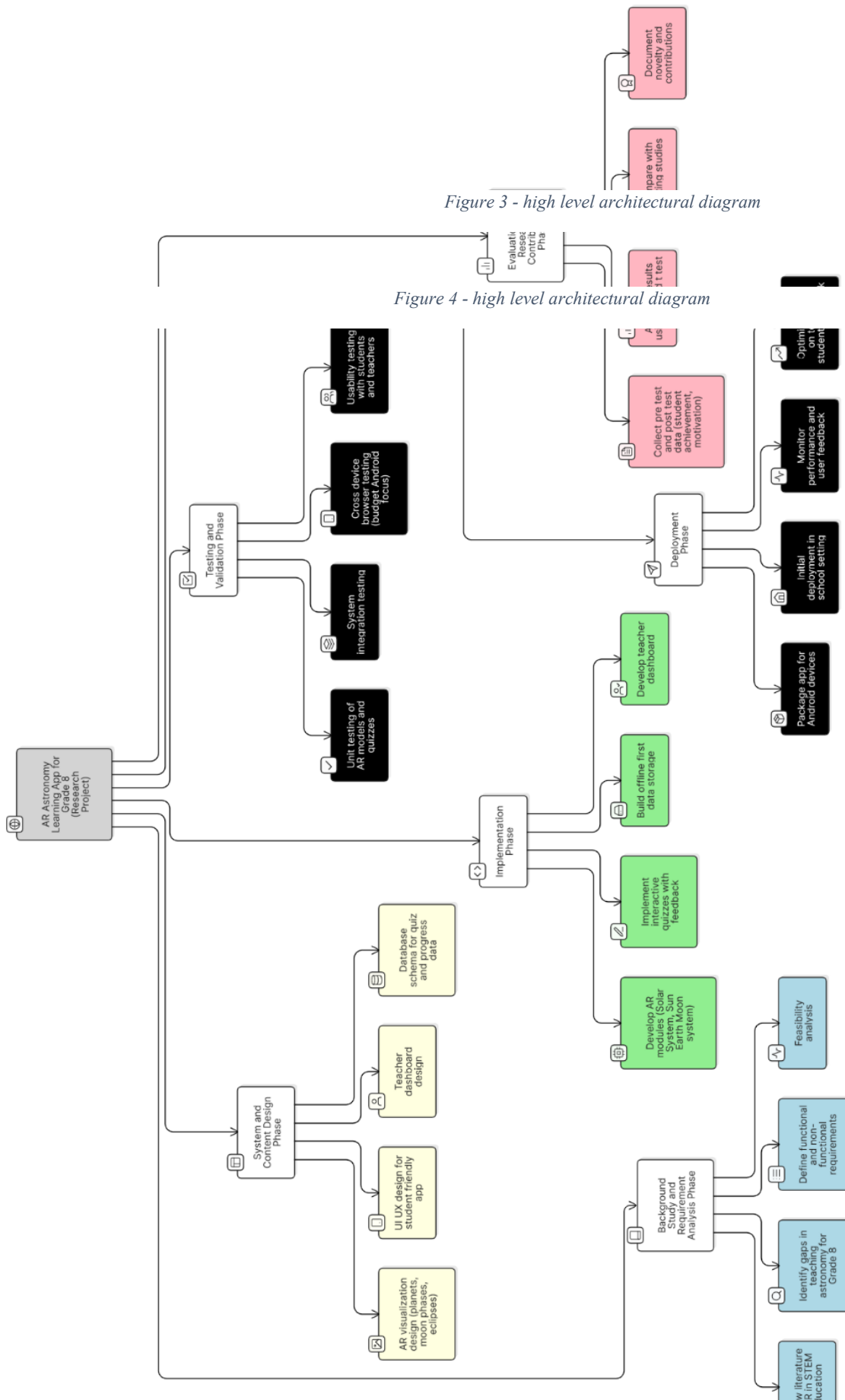


Figure 35- high lecture architecture design

Figure 4 - high level architectural diagram

Figure 3 - high level architectural diagram

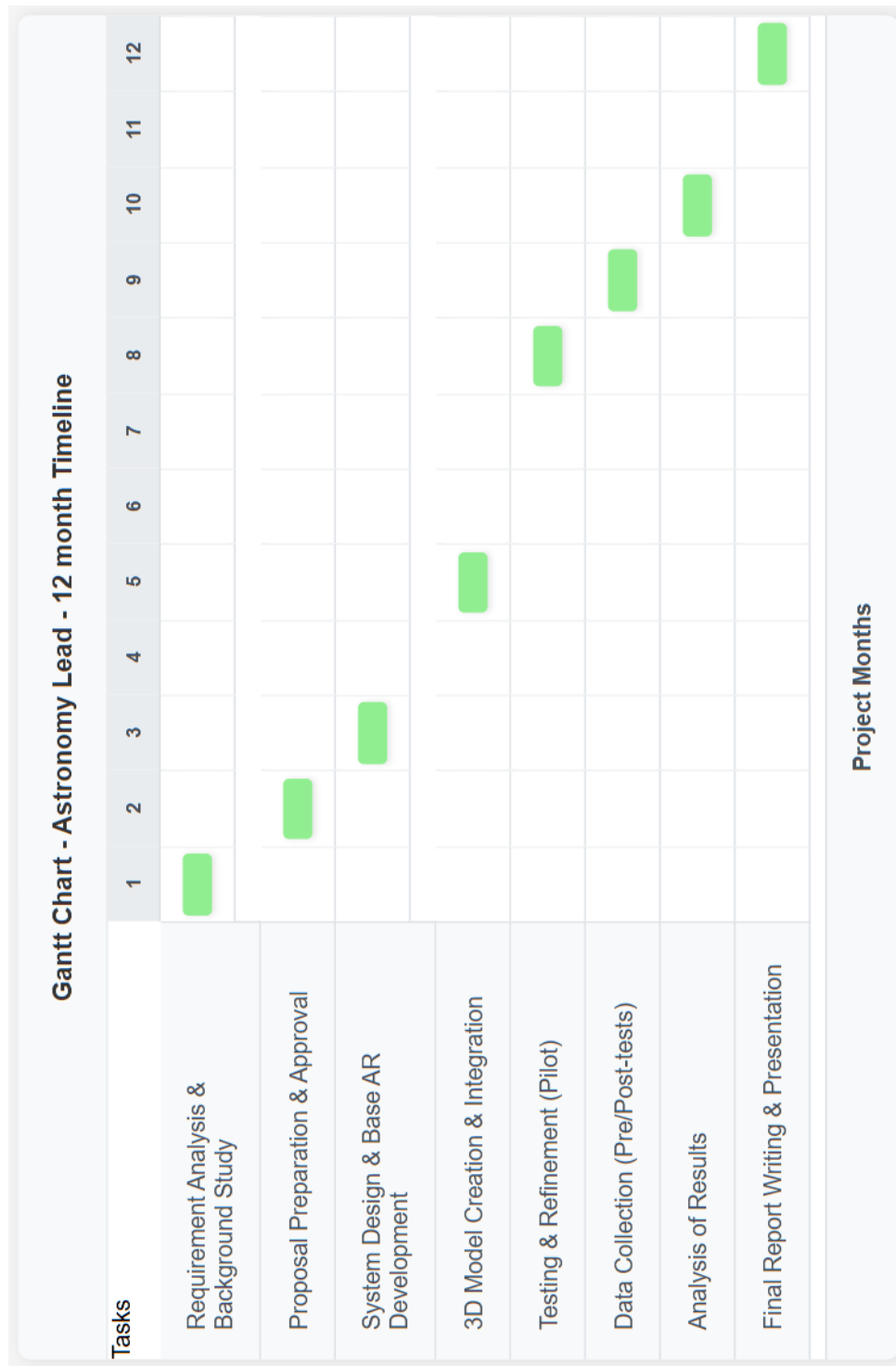


Table 2- gantt chart

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7. APPENDIX

- Sample teacher survey questionnaire.
- Sample student pre-test and post-test questions.
- Mockups of AR app interface screens.

